Solution

Bun5 × NON7

2110201: COMP ENG MATH (2023/2) Quiz 5

Name: Solution Student ID: 0123456789 Entry Number: 0

1. (5 points) Given a matrix $\mathbf{A} = \begin{array}{c} \begin{array}{cccc} p & p & p & f \end{array} \\ \begin{bmatrix} 1 & 0 & -2 & 0 \\ -2 & 1 & 3 & -2 \\ 0 & 0 & 1 & 2 \end{bmatrix} \end{array}$ $\rightarrow$ $RRE(A) = \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{bmatrix}$

1.1. (3 points) Find a basis of col($\mathbf{A}$). Also, find rank and nullity of $\mathbf{A}$.

Basis of col(A) = pivot columns of A

$= \left\{ \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix} \right\}$ #

rank(A) = dim (col(A)) = 3 #

nullity(A) = m − rank(A) = 4 − 3 = 1 #

1.2. (2 points) Find an implicit description of row($\mathbf{A}$). $\rightarrow$ write row(A) as perp of some vectors.

$row(A) = span\left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 3 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} \right\} = \boxed{perp\left\{ \begin{bmatrix} -4 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\} \#}$

then convert span to perp.

$\rightarrow \begin{bmatrix} 1 & 0 & -2 & 0 \\ -2 & 1 & 3 & -2 \\ 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{bmatrix}$ $\rightarrow$ horosenous generators $= \begin{bmatrix} -4 \\ 0 \\ -2 \\ 1 \end{bmatrix}$

2. (3 points, 1 point each) Provide short answer.

2.1. Find a basis for the orthocomplement of the subspace spanned by e_1, e_3, e_5 in $\mathbf{R}^6$. span$\{e_1, e_3, e_5\}$

(e_i is the standard basis vectors of $\mathbb{R}^6$)

$span\{e_1, e_3, e_5\}^{\perp} = perp\{e_1, e_3, e_5\} = nul \begin{bmatrix} - e_1 - \\ - e_3 - \\ - e_5 - \end{bmatrix} = span\{e_2, e_4, e_6\}$ #

$\therefore$ Basis is $\{e_2, e_4, e_6\}$ (no span!)

2.2. Find the nullity of $\mathbf{A}^T$ if A has a dimension of $m \times n$ and RRE($\mathbf{A}$) has k zero rows.

$m \left\{ \overbrace{\begin{bmatrix} 0 & \cdots & 0 \\ 0 & \cdots & 0 \end{bmatrix}}^{n} \right.$ $\left.\begin{matrix} \}m-k \\ \}k \end{matrix}\right.$ $m \times n$

$\rightarrow$ m − k non-zero rows $\rightarrow$ m − k pivots $\rightarrow$ rank(A) = n − k

$\therefore$ $rank(A^T) = m - k$ $\Rightarrow$ $nullity(A^T) = m - (m-k) = k$ #

(A^T has m columns)

2.3. Is the orthogonal complement of nul($\mathbf{A}$) spanned by the pivot columns of $\mathbf{A}^T$? Explain briefly.

Yes, $nul(A)^{\perp} = row(A) = col(A^T)$ and $col(A^T)$ is spanned by the pivot columns of A^T.

← you need to mention this for full score

(Note: row(A) is spanned by rows of RRE(A) and not A, hence the explanation not being completed.)

3. (2 points) Let $\mathbf{A}$ and $\mathbf{B}$ be a matrix with a dimension of 6×3 where col($\mathbf{A}$) is an orthogonal complement of col($\mathbf{B}$). Find the dimension of the left nullspace of $\mathbf{B}$.

$col(B) = col(A)^{\perp}$ — *

$dim(col(A)^{\perp}) + dim(col(A)) = 6$ $(col(A) \subset \mathbb{R}^6)$

$dim(col(B)) + dim(col(A)) = 6$

because both A and B have 3 columns $\rightarrow$ $dim(col(A)) \leq 3$ and $dim(col(B)) \leq 3$

$\therefore$ $dim(col(A)) = dim(col(B)) = 3$

$dim(nul(B^T)) = 6 - dim(col(B))$

$= 6 - 3 = 3$ #